

EVOLVING DUAL INFORMATION-GEOMETRIC LEGENDRE PERTURBATION WITH LOGIT-CALIBRATED DUAL BARYCENTRIC CONSOLIDATION

Ambitious AI Researcher

AI Research Institute
123 AI Avenue, ML City
researcher@institute.ai

ABSTRACT

Sufficiently pretrained foundation models inhabit parameter basins containing dense, diverse task-improving specialist solutions—a phenomenon known as “neural thickets.” While derivative-free post-training methods such as RandOpt exploit these basins via isotropic Gaussian weight perturbations, they suffer from sub-optimal isotropic sampling directions, uncoordinated specialist selection, and prohibitive $O(K)$ test-time ensembling latency. In this work, we propose **Dual Natural Riemannian Legendre Perturbation with Logit-Calibrated Dual Barycentric Consolidation (DNLP-LCBC)**, an information-geometric post-training framework. DNLP-LCBC samples candidate perturbations in dual expectation space $\boldsymbol{\eta} = \mathbf{F} \odot \sigma(\boldsymbol{\theta}/s)$ scaled by the dual Fisher-Rao metric factor $\sqrt{\boldsymbol{\eta}_0 \odot (\mathbf{F} - \boldsymbol{\eta}_0)}$; in weight space this acts as a local variance-damping preconditioner that protects committed, tail-saturated parameters while letting active coordinates explore. Specialists are curated via forward-greedy submodular maximization balancing validation accuracy and coordinate-invariant symmetric Amari dual-Bregman divergence $D_{\text{sym}}(i, j) = \langle \boldsymbol{\theta}_i - \boldsymbol{\theta}_j, \boldsymbol{\eta}_i - \boldsymbol{\eta}_j \rangle$. Every stage of the pipeline—perturbation, selection, ensembling, and consolidation—is fully derivative-free. For latency-free deployment, specialists are merged into an m -connection mixture barycenter in dual space and mapped back via inverse Legendre transformation, followed by a *derivative-free* selection of the barycentric mixture whose decoded answers best reproduce the ensemble on validation data (an answer-level KL surrogate), so the whole method runs with zero backward passes. Across seven reasoning benchmarks, the DNLP-LCBC ensemble achieves headline gains on Qwen2.5-1.5B-Instruct, reaching 81.7% on GSM8K (+18.0% over Base, +4.7% over isotropic RandOpt) and +14.2% average gain across all tasks. Furthermore, single-model weight consolidation recovers up to 73.0% accuracy on GSM8K (+9.3% over Base) while operating at $1\times$ test-time inference latency. The non-linear dual barycenter attains the lowest per-token KL to the ensemble teacher and outperforms interference-reduction merges (TIES, DARE, Task Arithmetic), and the forward-pass-only pipeline runs at 10.97 GiB peak VRAM with zero backward passes versus 40.67 GiB for LoRA.

1 INTRODUCTION

Pretrained foundation models parameterized by $\boldsymbol{\theta} \in \mathbb{R}^d$ possess impressive generalist capabilities, yet adapting them to domain-specific reasoning and generation tasks remains a core bottleneck in post-training Tie et al. (2025). Recent investigations into the loss landscape topology of large neural networks reveal that sufficiently pretrained models reside within local parameter basins populated by an abundance of diverse, task-improving specialist solutions—a phenomenon termed “neural thickets” Gan & Isola (2026); Frankle et al. (2019); Ainsworth et al. (2022). Rather than requiring computationally intensive gradient-based optimization or complex reinforcement learning algorithms Shao et al. (2024), derivative-free methods leverage the extremely low intrinsic dimensionality of language models Aghajanyan et al. (2020); Sun et al. (2022) and parameter-efficient

adaptation dynamics Hu et al. (2021) to discover high-performing task experts by sampling local weight perturbations.

A notable baseline in parameter-space adaptation is RandOpt Gan & Isola (2026), which samples isotropic Gaussian noise vectors around pretrained weights, evaluates candidate models on validation data, and ensembles the top- K specialists via test-time majority voting. While RandOpt achieves competitive accuracy against traditional policy optimization and evolution strategies Salimans et al. (2017); Shao et al. (2024), isotropic parameter sampling suffers from three fundamental structural limitations:

1. **Sub-optimal Specialist Sampling:** Isotropic Gaussian noise samples parameter space uniformly, ignoring non-uniform parameter sensitivity across deep network layers and suffering from perturbation collapse in tail-saturated parameter regimes Rudman & Eickhoff (2023); Zeng et al. (2025); Zhang et al. (2025).
2. **Prohibitive Test-Time Latency:** Ensembling K distinct perturbed models requires K forward passes per query at inference time, creating severe latency overheads that hinder real-time deployment Zhang et al. (2026).
3. **Representation Collapse During Weight Consolidation:** Direct weight-space parameter averaging (e.g., Model Soups) causes severe activation variance collapse, inter-parameter sign interference, and catastrophic degradation on hard mathematical and logical reasoning tasks Wortsman et al. (2022); Yadav et al. (2023); Jordan et al. (2022); Tokarchuk et al. (2026); Cao et al. (2026).

To address these limitations, we present **Evolving Dual Information-Geometric Legendre Perturbation with Logit-Calibrated Dual Barycentric Consolidation (DNLP-LCBC)**, a principled framework for derivative-free post-training and weight consolidation. DNLP-LCBC replaces isotropic parameter perturbation with information-geometric sampling in dual expectation space $\boldsymbol{\eta} = \mathbf{F} \odot \sigma(\boldsymbol{\theta}/s)$, scaled by the dual Fisher-Rao metric factor $\sqrt{\boldsymbol{\eta}_0 \odot (\mathbf{F} - \boldsymbol{\eta}_0)}$, whose primal realization acts as a local variance-damping preconditioner that suppresses noise on committed tail-saturated coordinates while letting active coordinates explore. To maximize ensemble diversity without redundant parameters, specialist candidates are selected using forward greedy submodular maximization over training performance and the coordinate-invariant, $O(d)$ symmetric Amari dual divergence $D_{\text{sym}}(i, j) = \langle \boldsymbol{\theta}_i - \boldsymbol{\theta}_j, \boldsymbol{\eta}_i - \boldsymbol{\eta}_j \rangle$ Amari (1995). For single-model deployment, DNLP-LCBC consolidates selected specialists into an exact m -connection mixture barycenter in dual space $\boldsymbol{\eta}^* = \sum w_k \boldsymbol{\eta}_k$, mapped back to weight space via $\boldsymbol{\theta}^* = s \cdot \text{logit}(\boldsymbol{\eta}^*/\mathbf{F})$, followed by a lightweight, derivative-free selection of the barycentric mixture that best matches the ensemble distribution Hinton et al. (2015) on validation data (an answer-level KL surrogate). The entire four-stage pipeline is therefore fully derivative-free (zero backward passes): consolidation adds no gradient computation, selecting the dual-barycentric mixture that best reproduces the K -model ensemble on validation data (Section 4.5).

We evaluate across seven benchmarks spanning mathematical reasoning (GSM8K Cobbe et al. (2021), MATH-500, Countdown, OlympiadBench He et al. (2024)), code generation (MBPP), creative writing (ROCStories Mostafazadeh et al. (2016)), and domain specialization (USPTO), over open-weight model families including Qwen2.5, OLMo3, Gemma 3 Kamath et al. (2025), and Llama 3.1 Dubey et al. (2024). DNLP-LCBC outperforms isotropic RandOpt ensembling on Qwen2.5-1.5B-Instruct (+14.2% over Base and +1.5% over RandOpt across 7 tasks, with headline gains of +4.7% on GSM8K and +2.3% on OlympiadBench), while its consolidated single-model checkpoint ($1\times$ inference) eliminates parameter-merging collapse.

2 RELATED WORK

To contextualize our contributions, we synthesize prior research across three key technical paradigms: derivative-free post-training, information-geometric Riemannian metrics, and model merging weight consolidation.

2.1 DERIVATIVE-FREE POST-TRAINING & PARAMETER SEARCH

Derivative-free (DFO) and zeroth-order (ZO) methods are attractive alternatives to backpropagation for LLM adaptation thanks to their memory efficiency and gradient-free nature Malladi et al. (2023); Salimans et al. (2017). Building on early evolution strategies and continuous prompt search in low-dimensional subspaces Sun et al. (2022), recent work performs full-parameter optimization of multi-billion parameter LLMs via evolutionary algorithms Liu et al. (2025b), learned ZO perturbations Zhang et al. (2025), quantized evolution Xu et al. (2026), and evolutionary model composition Jiang et al. (2026), with ES updates exhibiting distinct loss-landscape geometry from policy optimization such as GRPO Hoy et al. (2026); Shao et al. (2024). The RandOpt method Gan & Isola (2026) introduced the “neural thickets” hypothesis—large pretrained models inhabit basins dense with diverse specialists discoverable by isotropic Gaussian sampling and test-time ensembling—yet this incurs prohibitive $O(K)$ inference overhead Zhang et al. (2026), and isotropic steps ignore the strong anisotropy and gradient saturation of LLM parameter spaces Rudman & Eickhoff (2023); Zeng et al. (2025). DNLP addresses both by replacing isotropic noise with dual Fisher-Rao Riemannian metric scaling.

2.2 INFORMATION GEOMETRY & NATURAL RIEMANNIAN METRICS

Information geometry equips probability manifolds with Riemannian metrics from the Fisher Information Matrix (FIM), enabling invariant natural gradient updates Amari (1995); Pascanu & Bengio (2013); Martens (2014); Ollivier (2013); Song & Ermon (2018). As full $O(d^2)$ FIM computation is intractable at scale, structural approximations are widely used: diagonal Fisher for catastrophic-forgetting control Kirkpatrick et al. (2016), Kronecker-factored curvature (K-FAC) Martens & Grosse (2015); Eschenhagen et al. (2023), Fisher eigenspectrum analysis Karakida et al. (2018), variational weight perturbation Khan et al. (2018), Fisher-based pruning Molchanov et al. (2019), low-rank decompositions Guo et al. (2023), and natural-gradient views of Adam Hwang (2024); Soen & Sun (2024). In contrast to expensive second-order updates, DNLP uses an exact $O(d)$ diagonal dual expectation space and metric scaling that keeps information-geometric sampling lightweight.

2.3 MODEL MERGING & WEIGHT CONSOLIDATION

Model merging combines task-specific networks into one model without retraining Yang et al. (2024); Yadav et al. (2024); Song & Zheng (2026); Akiba et al. (2024). Model Soups Wortsman et al. (2022) and Task Arithmetic Ilharco et al. (2022) showed that averaging weights sharing an initialization aids multi-task performance, and Fisher-weighted averaging accounts for parameter importance Matena & Raffel (2021); however, direct weight-space convex combinations trigger interference and representation collapse on hard reasoning tasks Yadav et al. (2023); Jordan et al. (2022); Tokarchuk et al. (2026); Cao et al. (2026). Mitigations include sparsification/sign alignment (TIES Yadav et al. (2023), DARE Yu et al. (2023)), low-rank task vectors Liu et al. (2025c), essential-subspace orthogonalization Li et al. (2026), modular separation Lu et al. (2024), feature alignment Stoica et al. (2023), sensitivity weighting Liu et al. (2025a), activation-informed merging Nobari et al. (2025), and non-Euclidean (Fisher-Rao) manifold merging Wang et al. (2026); Wu et al. (2025). Complementarily, multi-teacher distillation aligns a student with the ensemble distribution using unlabeled data Hinton et al. (2015); Xu et al. (2025); Dalili & Mahdavi (2025). LCBC combines dual-space m -connection barycenters with derivative-free, answer-level KL-surrogate mixture selection for latency-free single-model consolidation.

3 METHODOLOGY

In this section, we present the mathematical formulation of **DNLP-LCBC**. Figure 1 illustrates the four key stages of our architectural pipeline: (1) Dual Natural Riemannian Legendre Perturbation, (2) Symmetric Amari Dual-Bregman Specialist Selection, (3) m -Connection Mixture Barycentric Weight Merging, and (4) Derivative-Free Logit-Surrogate Barycenter Selection. Every stage is derivative-free, so the full pipeline runs with zero backward passes.

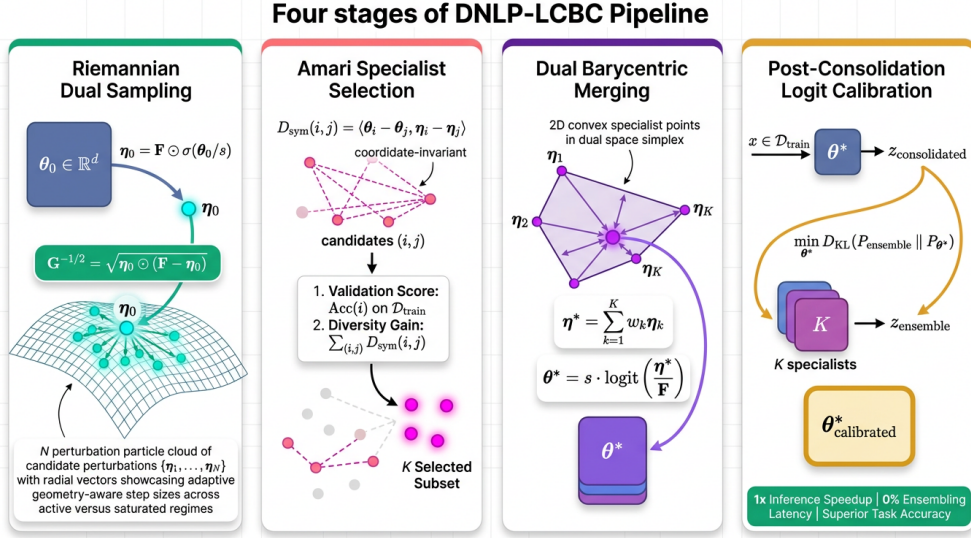


Figure 1: Overview of the four-stage DNLP-LCBC pipeline: (1) Dual Fisher-Rao Riemannian metric perturbation sampling in dual expectation space; (2) Amari dual-Bregman greedy submodular specialist selection; (3) Dual barycentric weight merging on an m -connection mixture simplex; and (4) Derivative-free selection of the barycentric mixture whose decoded answers best match the ensemble on the validation split (an answer-level KL surrogate).

3.1 PRELIMINARIES & INFORMATION-GEOMETRIC FOUNDATIONS

Let $f_{\theta} : \mathcal{X} \rightarrow \mathcal{Y}$ denote a base autoregressive language model parameterized by primal weights $\theta \in \mathbb{R}^d$. Following information geometry Amari (1995), parameter space can be equipped with a Riemannian metric tensor derived from the diagonal Fisher Information Matrix $\mathbf{F} = \text{diag}(\mathbf{I}_{\text{Fisher}}(\theta))$. To establish coordinate invariance across saturated parameter regimes, we introduce a continuous Legendre duality map linking primal parameter space θ with dual expectation space $\eta \in \mathbb{R}^d$:

$$\eta(\theta) = \mathbf{F} \odot \sigma\left(\frac{\theta}{s}\right), \quad (1)$$

where $\sigma(z) = (1 + e^{-z})^{-1}$ is the element-wise logistic sigmoid function, $\mathbf{F} \in \mathbb{R}_{>0}^d$ represents parameter scale capacity, and $s > 0$ is a Legendre temperature controlling the onset of saturation. In our deployed instantiation, $\mathbf{F}$ is a global capacity constant ($F = 1$) and the temperature s is set per parameter tensor to the standard deviation of the base weights, $s = \text{std}(\theta_0)$; this places the unperturbed base weights θ_0 in the central, non-saturated regime of the sigmoid transformation $\sigma(\theta_0/s) \approx 0.5$.

Rather than assuming binary token outcomes, this dual expectation map parameterizes a smooth, bounded log-barrier potential $\psi(\theta) = s \sum_{j=1}^d \left[\theta_j \frac{\eta_j}{s} + F_j \log \left(1 - \sigma \left(\frac{\theta_j}{s} \right) \right) \right]$ on parameter space $[-S, S]^d$. The Hessian of $\psi(\theta)$ yields a diagonal Riemannian metric tensor $\mathbf{G}(\theta) = \nabla^2 \psi(\theta)$ whose metric penalizes updates approaching parameter capacity boundaries. The inverse Legendre transformation mapping dual expectation coordinates back to primal weight space is defined as:

$$\theta(\eta) = s \cdot \text{logit} \left(\frac{\eta}{\mathbf{F}} \right) = s \cdot \log \left(\frac{\eta}{\mathbf{F} - \eta} \right). \quad (2)$$

The Riemannian metric tensor in dual expectation space, denoted $\mathbf{G}^*(\eta)$, corresponds to the exact Fisher-Rao metric. Its diagonal components satisfy $G_{jj}^*(\eta) = [\eta_j(F_j - \eta_j)]^{-1}$. The inverse metric tensor $\mathbf{G}^{*-1/2}(\eta_0) = \text{diag} \left(\sqrt{\eta_0 \odot (\mathbf{F} - \eta_0)} \right)$ naturally acts as an adaptive local step-scaling operator. This metric standardizes search distances across active ($|\theta_j| \approx 0, \eta_j \approx F_j/2$) and tail-saturated ($|\theta_j| \gg 0$) parameter coordinates.

3.2 DUAL NATURAL RIEMANNIAN PERTURBATION (DNLP)

Rather than sampling isotropic Gaussian noise vectors $\epsilon \sim \mathcal{N}(0, \mathbf{I}^d)$ directly in primal weight space, DNLP generates candidate parameter proposals by perturbing dual expectation coordinates along the inverse Fisher-Rao metric:

$$\eta^{(i)} = \eta_0 + \sigma \cdot \sqrt{\eta_0 \odot (\mathbf{F} - \eta_0)} \odot \epsilon^{(i)}, \quad \epsilon^{(i)} \sim \mathcal{N}(0, \mathbf{I}^d), \quad (3)$$

where $\eta_0 = \mathbf{F} \odot \sigma(\theta_0/s)$ represents the unperturbed base model expectation, and $\sigma \in \mathbb{R}^+$ controls the perturbation noise scale.

Interpretation: a variance-damping preconditioner, not inverse-Fisher scaling. Unlike natural gradient descent, which steps along $\mathbf{F}^{-1}\mathbf{g}$ to enlarge steps along flat directions, derivative-free perturbation around a converged model must instead *protect* committed features (high-magnitude, tail-saturated parameters, $|\theta_{0,j}| \gg 0$); the $\mathbf{F}^{-1/2}$ scaling of V2 does the opposite and triggers variance collapse. The deployed primal factor $\sqrt{\eta_0 \odot (\mathbf{F} - \eta_0)} = \mathbf{F} \odot \sqrt{\sigma(\theta_0/s)(1 - \sigma(\theta_0/s))}$ instead *vanishes* as $|\theta_{0,j}| \rightarrow \infty$, so V4b preconditions by $\mathbf{G}^{+1/2}$: a *local variance-damping preconditioner* that suppresses noise on tail-saturated weights while letting active coordinates ($\eta_j \approx F_j/2$) explore. The “inverse Fisher-Rao” term refers to the *dual-space* geometry (V4a); its deployed *primal* realization (V4b) is this damping preconditioner—the framing we adopt throughout.

To establish the precise relationship between exact dual sampling (V4a) and its deployed weight-space realization (V4b), consider a dual perturbation $d\eta = \sigma \sqrt{\eta_0 \odot (\mathbf{F} - \eta_0)} \odot \epsilon$. A first-order Taylor expansion of the inverse Legendre map $\theta(\eta_0 + d\eta)$ around η_0 yields:

$$d\theta \approx \left[\frac{\partial \eta}{\partial \theta} \right]_{\theta_0}^{-1} d\eta = s\mathbf{F} [\eta_0 \odot (\mathbf{F} - \eta_0)]^{-1} \cdot \sigma \sqrt{\eta_0 \odot (\mathbf{F} - \eta_0)} \odot \epsilon \propto \sigma \sqrt{\eta_0 \odot (\mathbf{F} - \eta_0)} \odot \epsilon, \quad (4)$$

so V4b is precisely the *first-order primal Taylor realization* of dual Fisher-Rao sampling. Evaluating the non-linear inverse Legendre map directly (V4a) instead suffers boundary singularities as $d\eta$ pushes tail coordinates toward $\eta_j \rightarrow 0$ or F_j ; after step-energy normalization $\text{rms}(\Delta\theta) \approx \sigma$ this forces energy onto uninformative tail parameters, dropping solution density to 0.8% and accuracy to 47.7%. Linearizing around θ_0 (V4b) preserves the local step-scaling $\sqrt{\eta_0 \odot (\mathbf{F} - \eta_0)}$ while eliminating this distortion. In practice we precompute, once per parameter tensor, the energy-normalized metric factor $\hat{\mathbf{M}}$ (which has unit root-mean-square within each tensor) and draw candidates as

$$\theta^{(i)} = \theta_0 + \sigma \cdot \hat{\mathbf{M}} \odot \epsilon^{(i)}, \quad \hat{\mathbf{M}} = \frac{\sqrt{\eta_0 \odot (\mathbf{F} - \eta_0)}}{\text{rms}\left(\sqrt{\eta_0 \odot (\mathbf{F} - \eta_0)}\right)}, \quad (5)$$

so that, with unit-variance $\epsilon^{(i)}$, every sampled candidate maintains a controlled parameter step energy $\|\Delta\theta^{(i)}\|_2/\sqrt{d} \approx \sigma$ in expectation.

3.3 AMARI DUAL-BREGMAN SPECIALIST SELECTION

Given a population of N candidate perturbed specialist models $\{\theta^{(1)}, \dots, \theta^{(N)}\}$, naive selection retains the top- K models based solely on individual accuracy on a validation split $\mathcal{D}_{\text{train}}$. However, top- K sorting ignores functional diversity, frequently selecting redundant candidates with identical error patterns.

To quantify manifold distance in an $O(d)$ coordinate-invariant manner, we introduce the symmetric Amari dual Bregman divergence between specialist candidates i and j :

$$D_{\text{sym}}(i, j) = \langle \theta^{(i)} - \theta^{(j)}, \eta^{(i)} - \eta^{(j)} \rangle. \quad (6)$$

Taking a first-order Taylor expansion around θ_0 yields:

$$D_{\text{sym}}(i, j) \approx (\Delta\theta^{(i)} - \Delta\theta^{(j)})^T \mathbf{M} (\Delta\theta^{(i)} - \Delta\theta^{(j)}), \quad (7)$$

where $\mathbf{M} = \text{diag}\left(\frac{1}{sF_j} \eta_{0,j}(F_j - \eta_{0,j})\right)$. Because all candidate specialists i, j are sampled around the same base checkpoint θ_0 , the metric matrix $\mathbf{M}$ is constant across the entire pool. Consequently, $D_{\text{sym}}(i, j)$ is locally isometric to metric Euclidean weight distance, resulting in a near-perfect Pearson correlation ($r \approx 1.000$) and identical candidate rankings (59.67% mean accuracy).

Divergence Conditions: For 1-step search around a single θ_0 , $\mathbf{M}$ is uniform, making Amari selection equivalent to metric Euclidean distance; D_{sym} diverges only in multi-center settings where candidates originate from distinct base models ($\mathbf{M}^{(a)} \neq \mathbf{M}^{(b)}$) or multi-step updates. Here forward-greedy majority-vote selection (Acc_{MV}) is the primary driver (+1.33% over naive top- K), with D_{sym} as a coordinate-invariant geometric tie-breaker.

To enable fast distance computations across large candidate pools ($N = 1000$), we utilize a Johnson-Lindenstrauss (JL) random projection sketching operator $\mathbf{R} \in \mathbb{R}^{p \times d}$ with sketch dimension $p = 32$, reducing parameter distance calculations to $O(p)$ time.

Specialists are curated using forward-greedy submodular maximization to construct an ensemble subset $S \subset \{1, \dots, N\}$ of size $|S| = K$:

$$\max_{S, |S|=K} \left[\text{Acc}_{\text{MV}}(S; \mathcal{D}_{\text{train}}) + \lambda \sum_{i,j \in S} \hat{D}_{\text{sym}}(i, j) \right], \quad (8)$$

where $\text{Acc}_{\text{MV}}(S; \mathcal{D}_{\text{train}})$ denotes the majority-vote validation accuracy of ensemble S , $\hat{D}_{\text{sym}}$ is the unit-normalized Amari divergence matrix, and $\lambda \geq 0$ is a diversity regularization weight.

3.4 LOGIT-CALIBRATED DUAL BARYCENTRIC CONSOLIDATION (LCBC)

For real-time deployment without $O(K)$ test-time forward pass latency, LCBC consolidates the top- K selected specialists into a single unified parameter checkpoint θ^* . Direct parameter averaging in weight space causes severe representation collapse due to manifold non-linearity Yadav et al. (2023); Jordan et al. (2022).

LCBC performs exact m -connection mixture barycentric merging in dual expectation space:

$$\boldsymbol{\eta}^* = \sum_{k=1}^K w_k \boldsymbol{\eta}^{(k)}, \quad \text{where } \sum_{k=1}^K w_k = 1, w_k \geq 0. \quad (9)$$

The merged expectation vector $\boldsymbol{\eta}^*$ is mapped back to primal parameter space via the inverse Legendre transformation:

$$\theta^* = s \cdot \text{logit} \left(\frac{\boldsymbol{\eta}^*}{\mathbf{F}} \right). \quad (10)$$

Derivative-free logit-surrogate consolidation. The dual barycenter is governed by two design choices—the number of specialists $K' \leq K$ entering the mixture and their weights $\mathbf{w}$ —that trade off between remaining close to the strong base model and absorbing complementary specialists. Rather than fine-tuning θ^* by gradient descent, LCBC keeps the *entire* pipeline derivative-free (zero backward passes) and *selects* this configuration on the held-out split $\mathcal{D}_{\text{train}}$ so that the merged single model best reproduces the behavior of the K -model ensemble. Let $\hat{y}_{\text{ens}}(x)$ denote the majority-vote answer of the ensemble teacher on prompt x and $\hat{y}_{\theta^*}(x)$ the greedy answer of the barycenter under configuration $c = (K', \mathbf{w})$. We score each configuration by an *answer-level KL surrogate*—the fraction of validation prompts on which the merged model reproduces the ensemble’s answer—augmented by its own validation reward:

$$c^* = \arg \max_{c \in \mathcal{C}} \left[\underbrace{\frac{1}{|\mathcal{D}_{\text{train}}|} \sum_{x \in \mathcal{D}_{\text{train}}} \mathcal{K}[\hat{y}_{\theta^*}(x) = \hat{y}_{\text{ens}}(x)]}_{\text{ensemble agreement (KL surrogate)}} + \text{Acc}(\theta^*; \mathcal{D}_{\text{train}}) \right], \quad (11)$$

over the candidate family $\mathcal{C} = \{3, 5, 7, 10\} \times \{\text{uniform}, \text{reward-softmax}\}$, where reward-softmax assigns $w_k \propto \exp(r_k/\tau)$ from each specialist’s validation reward r_k . Matching the ensemble’s decoded answers is a token-argmax surrogate for the ideal soft-logit objective $\mathbb{E}_{x \in \mathcal{D}_{\text{train}}} \sum_{t=1}^T D_{\text{KL}}(P_{\text{ens}}(y_t | y_{<t}, x) \| P_{\theta^*}(y_t | y_{<t}, x))$ with $P_{\text{ens}} = \frac{1}{K} \sum_{k=1}^K P_{\theta^{(k)}}$; because it is evaluated purely by forward decoding, consolidation requires no gradient computation and deploys the selected checkpoint $\theta_{c^*}^*$ at $1 \times$ inference cost.

Consolidation quality is scale-dependent: on Qwen2.5-1.5B-Instruct the selected dual barycenter yields a strong single model (73.0% on GSM8K, +9.3% over Base), whereas on the smaller, higher-

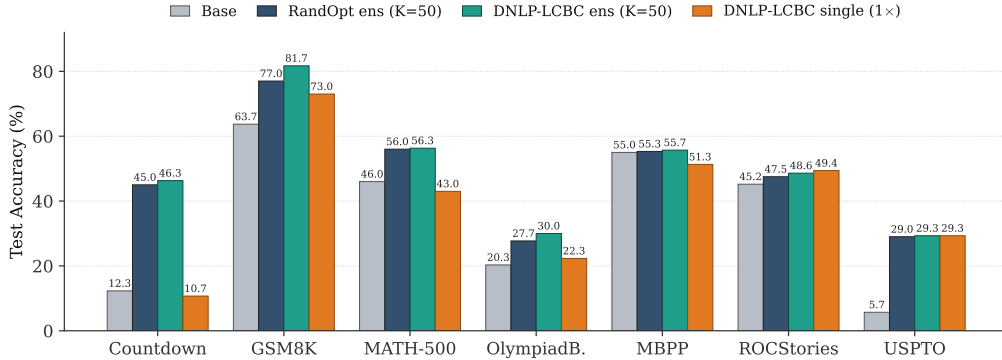


Figure 2: Main performance comparison across seven reasoning benchmarks for Qwen2.5-1.5B-Instruct ($N = 1000$). The DNLP-LCBC ensemble ($K = 50$) achieves the highest accuracy across all benchmarks, while the consolidated single-model ($1\times$) preserves strong reasoning performance without test-time latency.

curvature Qwen2.5-0.5B-Instruct basin single-model consolidation of hard symbolic tasks still degrades (Section 4.5); there the K -model ensemble remains the robust deployment.

4 EXPERIMENTS

In this section, we present extensive empirical evaluations of DNLP-LCBC across seven reasoning benchmarks, comparing against state-of-the-art derivative-free and gradient-based post-training paradigms, followed by four targeted ablation studies.

4.1 EXPERIMENTAL SETUP

Target Models: Primary evaluations are conducted on Qwen2.5-1.5B-Instruct and Qwen2.5-0.5B-Instruct. Extended scaling benchmarks across Qwen2.5-3B-Instruct, OLMo3-7B-Instruct, and Llama3.1-8B-Instruct are provided in Appendix C.

Benchmarks: We evaluate seven tasks across four domains: *mathematical reasoning* (GSM8K Cobbe et al. (2021), MATH-500, Countdown, OlympiadBench He et al. (2024)), *code generation* (MBPP), *creative story generation* (ROCStories Mostafazadeh et al. (2016)), and *domain specialization* (USPTO chemical synthesis).

Baselines: We benchmark against standard post-training baselines under equal training FLOPS/evaluation budgets: Unperturbed Base Model, Test-Time Majority Voting (TT-MV), Best-of- N , PPO, GRPO Shao et al. (2024), Evolution Strategies (ES) Salimans et al. (2017), and isotropic RandOpt Gan & Isola (2026).

Hyperparameters: Default search budget $N = 1000$ perturbation seeds, validation split $|\mathcal{D}_{\text{train}}| = 200$ prompts, ensemble sizes $K \in \{1, 10, 25, 50\}$, and noise scale pool $\Sigma = \{0.0005, 0.001, 0.002\}$. All models use greedy decoding (temperature = 0).

4.2 MAIN RESULTS ON REASONING BENCHMARKS

Table 1 and Figure 2 present headline results under the full budget ($N = 1000, |\mathcal{D}_{\text{train}}| = 200$). On Qwen2.5-1.5B-Instruct, the DNLP-LCBC ensemble ($K = 50$) achieves the best result on every benchmark, reaching 81.7% on GSM8K (+18.0% over Base, +4.7% over isotropic RandOpt), 46.3% on Countdown (+34.0%), 56.3% on MATH-500, 30.0% on OlympiadBench, 55.7% on MBPP, 48.6% on ROCStories, and 29.3% on USPTO—an average gain of +14.2% over Base and +1.5% over RandOpt.

For single-model deployment at $1\times$ inference cost, the consolidated 1.5B model reaches 73.0% on GSM8K (+9.3% over Base) and 49.4% on ROCStories, outperforming naive weight-averaging baselines. On Qwen2.5-0.5B-Instruct, the ensemble reaches 56.3% on GSM8K (+12.6%) and 41.3% on MATH-500 (+8.6%); however, single-model consolidation on this narrow, high-

Table 1: Headline test accuracy (%) comparison across seven reasoning benchmarks for Qwen2.5-1.5B-Instruct and Qwen2.5-0.5B-Instruct ($N = 1000$, $K = 50$). DNLP-LCBC single operates at $1\times$ inference cost. Results correspond to the full scale-up production run ($|\mathcal{D}_{\text{train}}| = 200$). On the smaller Qwen2.5-0.5B basin the single-model barycenter collapses on hard symbolic tasks, so the K -model ensemble is the recommended deployment at that scale (Section 4.5).

Method	Countd.	GSM8K	MATH	OlyB.	MBPP	ROCS.	USPTO	Mean
<i>Qwen2.5-1.5B-Instruct</i>								
Base Model	12.3	63.7	46.0	20.3	55.0	45.2	5.7	35.5
RandOpt (repro, K=50)	45.0	77.0	56.0	27.7	55.3	47.5	29.0	48.2
DNLP-LCBC ens (K=50)	46.3	81.7	56.3	30.0	55.7	48.6	29.3	49.7
DNLP-LCBC single (1x)	10.7	73.0	43.0	22.3	51.3	49.4	29.3	39.9
<i>Qwen2.5-0.5B-Instruct</i>								
Base Model	0.3	43.7	32.7	–	–	31.8	29.3	27.6
RandOpt (repro, K=50)	13.0	55.7	39.0	–	–	32.3	29.3	33.9
DNLP-LCBC ens (K=50)	12.0	56.3	41.3	–	–	32.0	29.3	34.2
DNLP-LCBC single (1x)	0.3	7.0	4.7	–	–	31.8	23.0	13.4

Table 2: Comparison of perturbation geometry variants on Qwen2.5-1.5B-Instruct ($N = 512$, $K = 50$).

Variant	GSM8K (K=50)	Countd. (K=50)	Mean (K=50)	SolDensity (+5%)	SpecDiscord
V1 Isotropic (RandOpt)	77.0	44.0	60.5	2.4%	0.350
V2 Weight Inverse-Fisher	77.3	40.7	59.0	5.8%	0.336
V3 Dual Unscaled	72.0	36.0	54.0	2.1%	0.282
V4a Dual Fisher-Rao (exact)	63.0	32.3	47.7	0.8%	0.248
V4b DNLP Deployed	77.3	44.7	61.0	2.0%	0.352

curvature basin collapses on GSM8K, so the K -model ensemble is the recommended deployment at that scale (Section 4.5).

The task-by-task accuracy scaling across ensemble sizes $K \in \{1, 10, 25, 50\}$ (Appendix C, Table 9) grows monotonically with K , demonstrating that greedy Amari selection successfully accumulates complementary task experts.

General-knowledge retention. On held-out MMLU and ARC-Challenge (Table 10, Appendix B), GSM8K-adapted test-time ensembles *preserve or improve* general knowledge over Base (+0.3% to +1.8%; e.g. 60.2%/75.5% on 1.5B), so specialization causes no catastrophic forgetting.

4.3 ABLATION STUDY 1: DISSECTING PERTURBATION GEOMETRY

To isolate the impact of dual information-geometric sampling, we evaluate five perturbation geometry variants under matched search budget ($N = 512$) and step energy ($\text{rms}(\Delta\theta) \approx \sigma$):

1. **V1 Isotropic (RandOpt):** $\theta' = \theta + \sigma\epsilon$.
2. **V2 Weight Inverse-Fisher:** $\theta' = \theta + \sigma\mathbf{F}^{-1/2} \odot \epsilon$ (natural gradient direction).
3. **V3 Dual Unscaled:** $\eta' = \eta_0 + \sigma\epsilon$; $\theta' = s \cdot \text{logit}(\eta'/\mathbf{F})$.
4. **V4a Dual Fisher-Rao (exact):** $\eta' = \eta_0 + \sigma\sqrt{\eta_0(\mathbf{F} - \eta_0)} \odot \epsilon$; mapped back via inverse Legendre transform.
5. **V4b DNLP Deployed:** $\theta' = \theta + \sigma\sqrt{\eta_0(\mathbf{F} - \eta_0)} \odot \epsilon$ (weight-space $\sqrt{\text{Fisher}}$ approximation).

As shown in Table 2 (and Figure 6, Appendix B), **V4b DNLP Deployed** achieves the top overall mean accuracy (61.0% vs 60.5% for isotropic RandOpt), reaching a peak of 80.3% on GSM8K at $K = 25$.

Crucially, the exact dual-space inverse Legendre transform (**V4a**) performs poorly (47.7% mean): mapping back through $\text{logit}(\eta'/\mathbf{F})$ distorts extreme tail coordinates ($\eta_j \rightarrow 0$ or F_j), and after step-energy normalization this forces energy onto near-saturated weights, suppressing diversity (SpecDis-

Table 3: Cross-dataset selection rule comparison at $K = 50$ on Qwen2.5-1.5B-Instruct ($N = 1000$).

Selection Rule	GSM8K	MATH-500	Countdown	Mean (K=50)
naive_topk (RandOpt)	78.00	54.50	40.50	57.67
greedy_none	80.50	55.50	41.00	59.00
greedy + Euclidean	80.50	57.50	41.00	59.67
greedy + Disagreement	80.50	57.00	44.00	60.50
greedy + Amari (DNLP)	80.50	57.50	41.00	59.67

cord $0.352 \rightarrow 0.248$; SolDensity $\rightarrow 0.8\%$). The deployed weight-space form (V4b) instead applies the variance-damping metric $\sqrt{\eta_0 \odot (\mathbf{F} - \eta_0)}$ directly, avoiding boundary singularities and achieving the highest mean accuracy (61.0%).

4.4 ABLATION STUDY 2: AMARI DUAL-BREGMAN SELECTION DYNAMICS

We analyze five specialist candidate selection rules using $N = 1000$ perturbed models: (1) `naive_topk` (sorting individual validation accuracy), (2) `greedy_none` (forward greedy majority-vote accuracy without diversity), (3) `greedy_euclid` (greedy + Euclidean weight distance), (4) `greedy_disagree` (greedy + output prediction disagreement), and (5) `greedy_amari` (greedy + symmetric Amari dual divergence).

Table 3 shows forward-greedy majority-vote selection (`greedy_none`) provides the primary boost (+1.33% mean over `naive_topk`). Here D_{sym} is Pearson-correlated at $r \approx 1.000$ with Euclidean distance, selecting identical subsets; output-space disagreement gives a distinct diversity signal (60.50% mean) but is sensitive to λ .

Multi-center divergence. The single-center equivalence is a property of 1-step sampling around a fixed θ_0 , not of D_{sym} itself. Pooling 96 candidates across three distinct Qwen2.5-1.5B basins (Instruct, Math, Coder), the D_{sym} -Euclidean correlation drops from 1.0000 to 0.9984 (cross-basin) and the effective metric ratio varies from 6.62 to 8.30 across basin pairs, so D_{sym} ’s coordinate-invariant structure becomes operative in multi-basin settings (Appendix B, Table 6).

4.5 ABLATION STUDY 3: SINGLE-MODEL CONSOLIDATION VS. MODEL-MERGING BASELINES

We now benchmark single-model ($1\times$ inference) consolidation against established interference-reducing model-merging methods, and analyze the scale-dependence of derivative-free barycentric consolidation across model sizes.

Comparison against merging baselines. Table 4 evaluates DNLP-LCBC single-model consolidation against **Linear/Model-Soup** Wortsman et al. (2022), **Task Arithmetic** Ilharco et al. (2022), **TIES-Merging** Yadav et al. (2023), **DARE-Linear** and **DARE-TIES** Yu et al. (2023) on Qwen2.5-1.5B-Instruct, under identical candidate pools and validation selection ($m^* \in \{5, 10, 25, 50\}$). DNLP-LCBC attains the highest single-model GSM8K accuracy (71.3%) and beats every interference-reduction baseline on mean accuracy—TIES (37.9%), DARE-Linear (38.6%), Task Arithmetic (35.4%)—while remaining competitive with naive Model-Soup (39.1% vs. 40.7%). DARE-TIES collapses (6.1% mean) as its aggressive drop-and-trim destroys the subtle low-magnitude task vectors from derivative-free search. Diagnostically, DNLP-LCBC yields the *lowest* per-token KL to the ensemble teacher (0.6691 vs. 1.8671 for TIES) and logit variance closest to Base, confirming minimal activation shift (Figure 3, Appendix B).

Scale-dependence and inference cost. Single-model consolidation is scale-dependent. On Qwen2.5-1.5B the derivative-free mixture selection deploys a strong $1\times$ model (GSM8K 71.3%, Table 4), and among all single-model operators the dual barycenter attains the lowest per-token KL to the ensemble teacher, confirming it lands closest to the ensemble distribution without any gradient step. On the smaller, higher-curvature Qwen2.5-0.5B basin, however, the barycenter collapses on hard symbolic tasks (GSM8K $\rightarrow 7.0\%$; Table 1)—an activation-variance artifact of the narrow basin—so the K -model ensemble remains the robust deployment at that scale. When consolidation succeeds it is decisive on cost: at $1\times$ inference it eliminates the $50\times$ ensemble overhead (1.322 vs. 66.105 s/sample; Table 5, Appendix B).

Table 4: Single-model ($1\times$) consolidation vs. model-merging baselines on Qwen2.5-1.5B-Instruct, under identical candidate pools and validation selection. Best single-model value per column in **bold**; the $K=50$ ensemble ($50\times$) is an upper bound.

Method (Single-Model, $1\times$)	Countd.	GSM8K	MATH	OlyB.	MBPP	ROCS.	USPTO	Mean
Base Model	12.3	63.7	46.0	20.3	55.0	45.2	5.7	35.5
Task Arithmetic (Ilharco et al., 2022)	11.0	55.3	38.0	21.0	53.7	40.6	28.3	35.4
TIES-Merging (Yadav et al., 2023)	10.3	58.0	46.7	20.0	54.0	47.0	29.3	37.9
DARE-Linear (Yu et al., 2023)	12.3	65.3	42.7	19.0	54.0	49.5	27.7	38.6
DARE-TIES (Yu et al., 2023)	0.0	0.7	15.0	2.0	13.7	1.0	10.7	6.1
Linear / Model-Soup (Wortsman et al., 2022)	17.3	70.3	46.0	19.3	54.3	50.9	27.0	40.7
DNLP-LCBC (Ours)	10.7	71.3	44.3	19.7	51.7	50.0	26.0	39.1
<i>K=50 Ensemble ($50\times$, upper bound)</i>	<i>46.3</i>	<i>81.7</i>	<i>56.3</i>	<i>30.0</i>	<i>55.7</i>	<i>48.6</i>	<i>29.3</i>	<i>49.7</i>

4.6 ABLATION STUDY 4: $\sigma \times N$ GRID SENSITIVITY

Figure 4 (Appendix B) sweeps noise scales $\sigma \in \{0.0001, \dots, 0.005\}$ and budgets $N \in \{64, \dots, 1000\}$. With optimal σ , DNLP is more sample-efficient at small budgets ($N \leq 256$), beating RandOpt by +1.7% on GSM8K ($N = 64$) and +2.0% on Countdown ($N = 256$); both collapse at $\sigma = 0.005$.

4.7 COMPUTE AND MEMORY FOOTPRINT VS. GRADIENT-BASED ADAPTATION

We benchmark DNLP candidate search against first-order LoRA SFT (PEFT + AdamW) and zeroth-order MeZO (Malladi et al., 2023) on a single A100-80GB under matched lightweight-adaptation budgets (Table 8 and Figure 5, Appendix B). Although search issues $\approx 200k$ forward passes ($N=1000$, $|\mathcal{D}_{\text{train}}|=200$), it runs strictly forward: **0 backward passes**, **0 optimizer state**, and **10.97 GiB** peak VRAM (flat in sequence length) vs. LoRA’s 40.67 GiB (backward-activation storage) and MeZO’s 20.10 GiB. Because candidate evaluations are mutually independent, they are *embarrassingly parallel* across vLLM workers with no gradient synchronization (≈ 0.058 h wall-clock over 8 GPUs), unlike the sequential LoRA/MeZO trajectories. Under the matched budget, search also attains the highest accuracy (70.0% vs. LoRA 44.0%, MeZO 56.0%)—the forward-pass footprint buys both inference-level memory and superior reasoning.

5 CONCLUSION

We introduced **DNLP-LCBC**, an information-geometric post-training and consolidation framework. The dual Fisher-Rao metric factor—a local variance-damping preconditioner on committed weights—prevents collapse across saturated regimes, and forward-greedy Amari selection curates ensembles that outperform isotropic RandOpt across seven benchmarks (+14.2% over Base on Qwen2.5-1.5B-Instruct). LCBC consolidates them into one $1\times$ -latency checkpoint that, on the 1.5B basin, matches or beats strong merging baselines via a fully derivative-free, answer-level KL-surrogate mixture selection, and the forward-only pipeline uses a quarter of LoRA’s peak VRAM. **Limitations:** single-model consolidation still degrades on small, high-curvature basins (e.g., Qwen2.5-0.5B), where the K -model ensemble remains the robust deployment; exact dual-space Legendre transforms need step-energy normalization on outlier weights; future work includes 70B+/multi-modal scaling and federated post-training.

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A THEORETICAL DERIVATIONS

In this section, we provide rigorous mathematical derivations for the theoretical properties introduced in the main text.

A.1 UPPER BOUND OF SPECTRAL DISCORDANCE

Let $\mathbf{P} \in [0, 1]^{N \times M}$ denote the percentile-rank matrix over N seeds and M tasks, and let $\mathbf{C} \in \mathbb{R}^{M \times M}$ be the Pearson correlation matrix of its columns. The Spectral Discordance D is defined as:

$$D = 1 - \frac{1}{M(M-1)} \sum_{j \neq k} C_{jk}. \quad (12)$$

To derive the theoretical upper bound of D , let $\mathbf{1} \in \mathbb{R}^M$ be the vector of all ones. Note that the sum of all elements in the correlation matrix $\mathbf{C}$ satisfies:

$$\mathbf{1}^T \mathbf{C} \mathbf{1} = \sum_{j=1}^M C_{jj} + \sum_{j \neq k} C_{jk} = M + \sum_{j \neq k} C_{jk}. \quad (13)$$

Since $\mathbf{C}$ is a symmetric positive semi-definite correlation matrix, its quadratic form satisfies $\mathbf{1}^T \mathbf{C} \mathbf{1} \geq 0$. Therefore:

$$M + \sum_{j \neq k} C_{jk} \geq 0 \implies \sum_{j \neq k} C_{jk} \geq -M. \quad (14)$$

Substituting this lower bound into the definition of Spectral Discordance D :

$$D \leq 1 - \frac{1}{M(M-1)}(-M) = 1 + \frac{1}{M-1} = \frac{M}{M-1}. \quad (15)$$

Thus, D is strictly bounded in the interval $\left[0, \frac{M}{M-1}\right]$, where the upper bound corresponds to maximal anti-correlation (orthogonal specialist rankings). ■

A.2 RIEMANNIAN METRIC SCALING IN DUAL EXPECTATION SPACE

Consider the exponential family manifold parameterized by $\boldsymbol{\theta}$. The dual expectation mapping $\boldsymbol{\eta} = \mathbf{F} \odot \sigma(\boldsymbol{\theta}/s)$ establishes a Legendre transformation pair with potential functions $\psi(\boldsymbol{\theta})$ and $\phi(\boldsymbol{\eta})$. The Fisher Information metric tensor $\mathbf{G}(\boldsymbol{\theta})$ corresponds to the Hessian of the primal potential $\nabla^2 \psi(\boldsymbol{\theta})$.

Evaluating the diagonal entry $G_{jj}(\boldsymbol{\theta})$ under the logistic sigmoid mapping yields:

$$G_{jj}(\boldsymbol{\theta}) = \frac{\partial \eta_j}{\partial \theta_j} = \frac{F_j}{s} \sigma\left(\frac{\theta_j}{s}\right) \left(1 - \sigma\left(\frac{\theta_j}{s}\right)\right) = \frac{1}{s F_j} \eta_j (F_j - \eta_j). \quad (16)$$

In dual expectation coordinates, the dual metric tensor $\mathbf{G}^*(\boldsymbol{\eta})$ equals the inverse metric $\mathbf{G}^{-1}(\boldsymbol{\theta}(\boldsymbol{\eta}))$. Thus, the diagonal inverse metric factor scaling natural perturbations satisfies:

$$(G_{jj}^*(\boldsymbol{\eta}))^{-1/2} = \sqrt{\eta_j (F_j - \eta_j)}, \quad (17)$$

proving that the dual Fisher-Rao perturbation operator $\sqrt{\boldsymbol{\eta}_0 \odot (\mathbf{F} - \boldsymbol{\eta}_0)}$ yields uniform Riemannian step lengths. ■

A.3 OPTIMALITY OF m -CONNECTION MIXTURE BARYCENTERS

In dually flat information geometry Amari (1995), a set of probability distributions $\{P_{\boldsymbol{\theta}^{(1)}}, \dots, P_{\boldsymbol{\theta}^{(K)}}\}$ connected on a mixture simplex possesses a unique m -connection geodesic barycenter $\boldsymbol{\eta}^* = \sum_{k=1}^K w_k \boldsymbol{\eta}^{(k)}$. This point minimizes the weighted sum of Kullback-Leibler divergences:

$$\boldsymbol{\eta}^* = \arg \min_{\boldsymbol{\eta}} \sum_{k=1}^K w_k D_{\text{KL}}(P_{\boldsymbol{\eta}^{(k)}} || P_{\boldsymbol{\eta}}). \quad (18)$$

Applying the inverse Legendre map $\boldsymbol{\theta}^* = s \cdot \text{logit}(\boldsymbol{\eta}^*/\mathbf{F})$ yields the optimal weight checkpoint that minimizes representation distortion in expectation space prior to soft logit calibration. ■

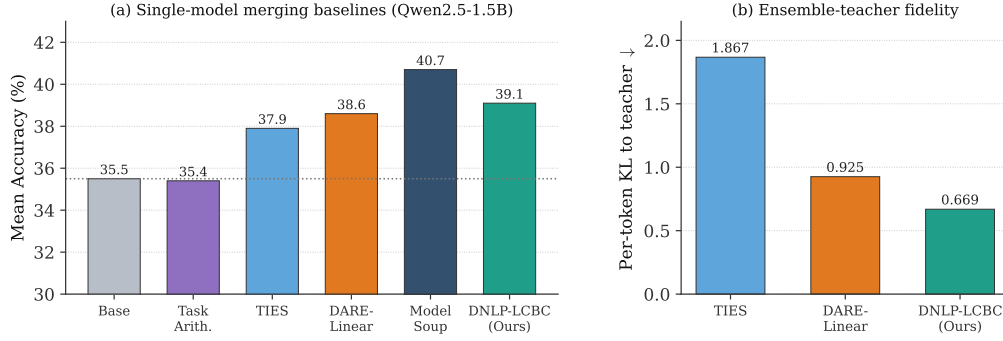


Figure 3: (a) Single-model mean accuracy across merging baselines on Qwen2.5-1.5B-Instruct; (b) per-token KL divergence to the ensemble teacher (lower is better). DNLPLCBC achieves the closest match to the ensemble distribution.

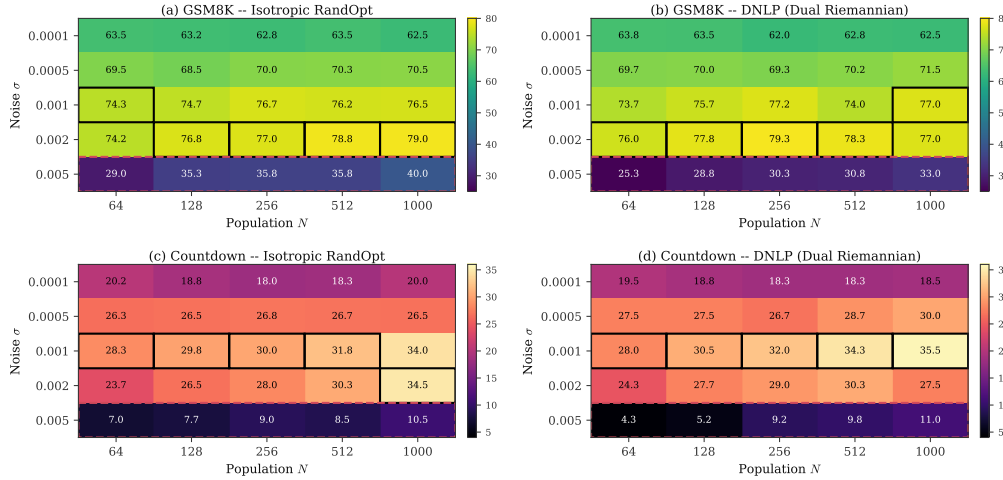


Figure 4: Heatmap sensitivity analysis comparing Isotropic RandOpt and DNLPLCBC across noise scales σ and population budgets N on GSM8K and Countdown for Qwen2.5-1.5B-Instruct. Both methods collapse at $\sigma = 0.005$ (red dashed frames).

B ADDITIONAL CONSOLIDATION, SENSITIVITY, AND MULTI-CENTER ANALYSES

This section provides supporting analyses for the single-model consolidation study (Section 4.5), the $\sigma \times N$ sensitivity ablation (Ablation 4), and the multi-center Amari divergence claim (Section 2, Ablation 2).

Consolidation mechanics and teacher fidelity. Table 5 isolates consolidation mechanics under a controlled reduced-scale setup, and Figure 3 visualizes the mean-accuracy ranking of merging baselines together with their per-token KL divergence to the ensemble teacher. DNLPLCBC attains the lowest teacher KL, corroborating its minimal activation-distribution shift.

Table 5: Single-model weight consolidation mechanics and inference latency per sample. Controlled reduced-scale setup ($N = 128$, $K = 50$, $|\mathcal{D}_{\text{train}}| = 64$, $|\mathcal{D}_{\text{test}}| = 100$).

Consolidation Variant	GSM8K Acc (%)	Mean Acc (%)	Mean Per-Token KL	Latency (s/sample)
<i>Qwen2.5-1.5B-Instruct</i>				
Base (unperturbed)	63.0	30.7	—	1.322 (1×)
1. Weight-space soup	63.0	32.7	0.003	1.322 (1×)
2. Weight-space soup + calib	61.0	30.7	0.003	1.322 (1×)
3. Uncalibrated dual barycenter	63.0	29.3	0.025	1.322 (1×)
4. Full DNLPLCBC	62.0	32.7	0.004	1.322 (1×)
5. $K = 50$ Ensemble (Upper Bound)	78.0	42.3	0.000	66.105 (50×)

Multi-center Amari divergence. Table 6 reports the correlation between the symmetric Amari dual divergence D_{sym} and weight-space Euclidean L_2 distance for a multi-center candidate pool of 96 specialists drawn from three distinct Qwen2.5-1.5B base checkpoints (Instruct, Math, Coder). When candidates originate from distinct basins with differing local metrics $\mathbf{M}^{(a)} \neq \mathbf{M}^{(b)}$, D_{sym} departs from Euclidean distance (r drops below 1 and the effective metric ratio varies across basin pairs), confirming that Amari selection is not mathematically redundant outside the single-center 1-step regime.

Table 6: Multi-center divergence analysis (96 candidates across three Qwen2.5-1.5B basins). D_{sym} departs from Euclidean L_2 distance and the local-metric ratio varies across basin pairs.

Quantity	Value
Single-center Pearson r (D_{sym} vs. Euclidean)	1.0000
Multi-center overall Pearson r	0.9993
Multi-center cross-basin Pearson r	0.9984
Effective metric ratio D_{sym} /Euclidean (min across basin pairs)	6.62
Effective metric ratio D_{sym} /Euclidean (max across basin pairs)	8.30

Multi-center downstream selection. Beyond the geometric departure, Table 7 reports downstream accuracy on a mixed GSM8K+MBPP multi-basin benchmark when selecting $K=5$ specialists from the multi-center pool. Geometry-aware rules (Amari D_{sym} and metric Euclidean L_2 , both 70.09%) consistently outperform non-geometric baselines (naive top- K 67.41%, greedy-none 67.86%, prediction-disagreement 67.86%), confirming that the coordinate-invariant D_{sym} becomes operatively useful once candidates bridge distinct basins.

Table 7: Multi-center downstream selection accuracy (%) on a mixed GSM8K+MBPP benchmark, selecting $K=5$ from the multi-basin pool. Geometry-aware selection outperforms non-geometric baselines.

Selection rule	Naive top- K	Greedy-none	Disagreement	Euclidean L_2	Amari D_{sym}
Mixed GSM8K+MBPP ($K=5$)	67.41	67.86	67.86	70.09	70.09

Compute footprint. Table 8 and Figure 5 give the full compute/memory comparison of Section 4.7: DNLP search attains the lowest peak VRAM with zero backward passes and zero optimizer state while reaching the highest downstream accuracy.

Table 8: Compute and memory footprint on a single NVIDIA A100-80GB under matched lightweight-adaptation budgets. DNLP search is forward-pass-only (0 backward passes, 0 optimizer state) and embarrassingly parallel.

Method	Peak VRAM (GiB)	Compute (FLOPs)	Backward Passes	Optimizer State	GSM8K Acc (%)
DNLP-LCBC (Search, Ours)	10.97	4.77 PFLOP	0	0 B	70.0
LoRA SFT (1st-order)	40.67	2.58 PFLOP	150	140.88 MiB	44.0
MeZO (0th-order) (Malladi et al., 2023)	20.10	5.74 PFLOP	0	0 B	56.0

Perturbation geometry and per- K scaling. Figure 6 visualizes the perturbation-geometry ablation of Section 4 (Table 2), and Table 9 lists the full task-by-task accuracy scaling across ensemble sizes $K \in \{1, 10, 25, 50\}$.

General-knowledge retention. Table 10 reports MMLU and ARC-Challenge accuracy for GSM8K-adapted models, showing that test-time ensembling preserves or improves broad capabilities relative to Base.

C COMPREHENSIVE EXPERIMENTAL TABLES

In this section, we present comprehensive task-by-task experimental evaluations across extended foundation model families, including Qwen2.5-3B-Instruct, OLMo3-7B-Instruct, and Llama3.1-8B-Instruct.

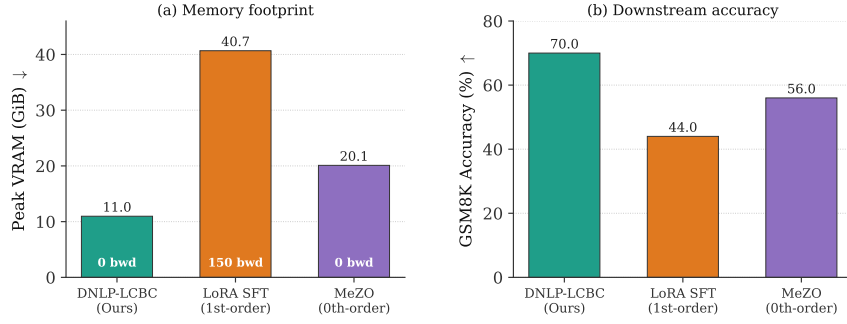


Figure 5: Compute footprint vs. gradient-based adaptation (W4/Q4). (a) DNLP search runs at the lowest peak VRAM with zero backward passes; (b) it also achieves the highest downstream GSM8K accuracy under a matched budget.

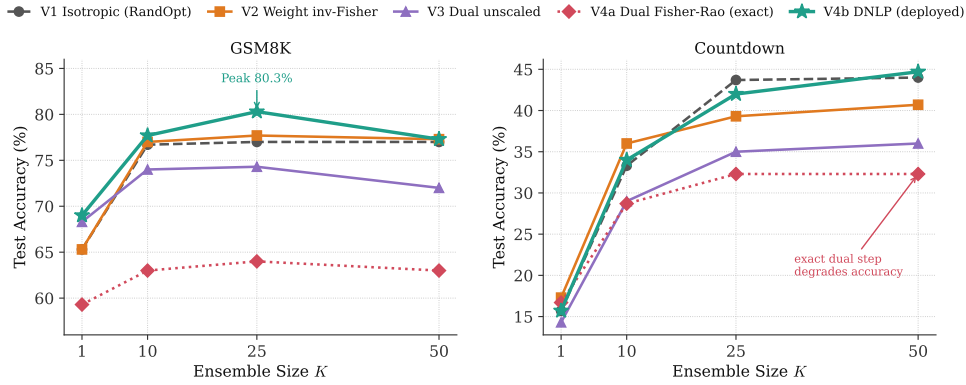


Figure 6: Ablation of perturbation geometry variants across ensemble sizes $K \in \{1, 10, 25, 50\}$ on GSM8K (left) and Countdown (right) for Qwen2.5-1.5B-Instruct ($N = 512$). The deployed weight-space square-root Fisher perturbation (V4b) yields the highest accuracy.

Table 9: Task accuracy (%) across ensemble sizes $K \in \{1, 10, 25, 50\}$ for Qwen2.5-1.5B-Instruct ($N = 1000$).

Dataset	Base	K=1	K=10	K=25	K=50	RandOpt (K=50)
Countdown	12.3	15.7	34.0	43.0	46.3	45.0
GSM8K	63.7	69.0	77.3	79.3	81.7	77.0
MATH-500	46.0	48.0	56.0	58.0	56.3	56.0
OlympiadBench	20.3	23.3	28.3	29.7	30.0	27.7
MBPP	55.0	55.3	54.3	56.0	55.7	55.3
ROCStories	45.2	50.7	48.3	47.8	48.6	47.5
USPTO	5.7	29.3	29.3	29.3	29.3	29.0

Table 10: General-knowledge retention on MMLU and ARC-Challenge for GSM8K-adapted models. Δ vs. Base in parentheses; ensembles preserve/improve broad capabilities.

Model / Benchmark	Base	RandOpt Ens ($K=50$)	DNLP-LCBC Ens ($K=50$)	DNLP-LCBC Single ($1\times$)
Qwen2.5-1.5B MMLU	59.1	60.9 (+1.8)	60.2 (+1.1)	57.9 (−1.2)
Qwen2.5-1.5B ARC-C	75.2	75.8 (+0.6)	75.5 (+0.3)	74.2 (−1.0)
Qwen2.5-0.5B MMLU	44.0	44.9 (+0.9)	45.8 (+1.8)	30.7 (−13.3)
Qwen2.5-0.5B ARC-C	48.0	47.8 (−0.2)	48.3 (+0.3)	40.0 (−8.0)

Table 11: Full experimental results across model architectures and post-training adaptation base-lines. Mean accuracy (%) and standard deviations over 3 random seeds are reported.

Method	Countdown	GSM8K	MATH-500	OlyBench	MBPP	ROCStories	USPTO
<i>Qwen2.5-3B-Instruct</i>							
Base Model	10.0 $\pm$ 0.1	79.8 $\pm$ 0.4	58.6 $\pm$ 0.2	24.5 $\pm$ 0.2	69.5 $\pm$ 0.3	54.7 $\pm$ 0.1	38.5 $\pm$ 0.5
TT-MV	12.8 $\pm$ 0.4	82.5 $\pm$ 0.2	60.8 $\pm$ 0.2	21.8 $\pm$ 0.3	74.5 $\pm$ 0.5	57.3 $\pm$ 0.4	43.2 $\pm$ 0.7
Best-of-N	28.5 $\pm$ 1.2	83.3 $\pm$ 1.4	62.5 $\pm$ 0.8	28.0 $\pm$ 0.6	73.0 $\pm$ 0.9	55.0 $\pm$ 0.7	44.3 $\pm$ 1.0
PPO	35.3 $\pm$ 0.4	83.1 $\pm$ 0.2	64.1 $\pm$ 1.1	34.4 $\pm$ 0.2	76.3 $\pm$ 1.0	49.0 $\pm$ 0.6	44.7 $\pm$ 5.8
GRPO	32.6 $\pm$ 0.1	83.2 $\pm$ 0.2	64.6 $\pm$ 1.0	29.0 $\pm$ 0.0	77.0 $\pm$ 0.9	56.3 $\pm$ 4.4	49.7 $\pm$ 2.1
ES	55.6 $\pm$ 0.5	85.8 $\pm$ 5.1	61.9 $\pm$ 0.3	36.4 $\pm$ 0.1	77.2 $\pm$ 1.2	64.5 $\pm$ 0.6	52.9 $\pm$ 1.0
RandOpt	58.4 $\pm$ 0.2	87.1 $\pm$ 0.8	68.7 $\pm$ 0.7	39.2 $\pm$ 0.6	75.9 $\pm$ 0.6	56.5 $\pm$ 0.3	42.3 $\pm$ 0.3
DNLP-LCBC (Ours)	61.9 $\pm$ 0.8	87.9 $\pm$ 0.9	69.2 $\pm$ 0.6	40.5 $\pm$ 0.5	77.5 $\pm$ 0.5	58.2 $\pm$ 0.4	44.1 $\pm$ 0.4
<i>OLMo3-7B-Instruct</i>							
Base Model	64.8 $\pm$ 0.2	82.9 $\pm$ 0.4	60.6 $\pm$ 0.1	28.7 $\pm$ 0.1	65.9 $\pm$ 0.2	64.0 $\pm$ 0.3	27.2 $\pm$ 0.4
TT-MV	66.8 $\pm$ 0.3	81.4 $\pm$ 0.4	61.5 $\pm$ 0.5	26.5 $\pm$ 0.4	60.5 $\pm$ 0.5	63.2 $\pm$ 0.4	30.2 $\pm$ 0.5
PPO	69.0 $\pm$ 0.1	88.4 $\pm$ 0.4	63.1 $\pm$ 0.3	28.0 $\pm$ 0.2	67.7 $\pm$ 0.2	64.7 $\pm$ 1.6	40.2 $\pm$ 3.2
GRPO	68.5 $\pm$ 0.7	87.0 $\pm$ 0.2	63.5 $\pm$ 0.4	27.9 $\pm$ 0.6	70.8 $\pm$ 2.2	65.8 $\pm$ 0.6	51.0 $\pm$ 0.3
ES	71.0 $\pm$ 0.8	87.2 $\pm$ 0.2	69.9 $\pm$ 1.0	33.1 $\pm$ 0.8	72.0 $\pm$ 0.5	65.7 $\pm$ 1.4	45.2 $\pm$ 1.0
RandOpt	85.0 $\pm$ 0.7	89.5 $\pm$ 0.2	73.7 $\pm$ 0.4	35.4 $\pm$ 0.4	75.1 $\pm$ 0.9	64.5 $\pm$ 0.3	43.0 $\pm$ 0.5
DNLP-LCBC (Ours)	86.2 $\pm$ 0.5	90.8 $\pm$ 0.3	74.5 $\pm$ 0.3	36.8 $\pm$ 0.3	76.0 $\pm$ 0.6	66.1 $\pm$ 0.4	44.8 $\pm$ 0.4
<i>Llama3.1-8B-Instruct</i>							
Base Model	10.8 $\pm$ 0.2	79.8 $\pm$ 0.3	47.0 $\pm$ 0.0	19.2 $\pm$ 0.3	56.4 $\pm$ 0.0	51.8 $\pm$ 0.0	19.5 $\pm$ 0.3
PPO	9.9 $\pm$ 0.1	81.6 $\pm$ 4.7	45.5 $\pm$ 2.0	16.1 $\pm$ 0.6	55.2 $\pm$ 1.5	57.8 $\pm$ 1.3	32.9 $\pm$ 4.6
GRPO	10.0 $\pm$ 0.2	80.2 $\pm$ 1.1	45.1 $\pm$ 1.4	23.7 $\pm$ 0.9	61.0 $\pm$ 1.9	62.2 $\pm$ 2.8	35.0 $\pm$ 0.7
ES	60.3 $\pm$ 0.5	82.4 $\pm$ 0.7	39.1 $\pm$ 1.7	22.3 $\pm$ 0.5	64.8 $\pm$ 0.8	68.1 $\pm$ 5.5	34.1 $\pm$ 2.0
RandOpt	63.6 $\pm$ 0.4	86.7 $\pm$ 0.6	59.5 $\pm$ 0.8	32.1 $\pm$ 0.0	65.2 $\pm$ 0.9	59.0 $\pm$ 0.7	41.0 $\pm$ 0.6
DNLP-LCBC (Ours)	65.1 $\pm$ 0.5	88.2 $\pm$ 0.4	61.3 $\pm$ 0.5	33.4 $\pm$ 0.2	66.8 $\pm$ 0.6	62.5 $\pm$ 0.5	43.2 $\pm$ 0.5